



Optimization Techniques for Multi Cooperative Systems MCTR 1021
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This is to certify that:

- (i) the report comprises only my original work toward the course project,
- (ii) due acknowledgment has been made in the text to all other material used

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Acknowledgments

I would like to thank the following for their efforts with me in my study and for tolerating me during this journey...

Abstract

The abstract of the document is added here...

It is usually written after finishing writing all the other chapters.

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List of Abbreviations

MRSs	Multi-Robot Systems
UAVs	Unmanned Aerial Vehicles
UGVs	Unmanned Ground Vehicles

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Chapter 1

Introduction

1.1 Section Name

Some sample text with an Unmanned Aerial Vehicles (UAVs), some citation [?, ?], and some more Unmanned Ground Vehicles (UGVs).

1.2 Another Section

Reference to Section 1.1, and reuse of UGVs nad UAVs with also full use of Multi-Robot Systems (MRSs). Reference to figure 1.1.



Figure 1.1: GUC Logo

Chapter 2

Literature Review

Chapter 3

Methodology

Problem Formulation

Assumptions

The environment is a grid map $M \in \{0, 1, 2\}^{n \times m}$ with unit cells 1×1 ; $M[x, y] = 1$ means free, 0 means obstacle, and 2 marks a robot's position. Obstacles are unknown a priori. As robots explore, newly detected obstacles are shared; let O_t be the set known by time t , with $O_t \subseteq O_{t+1}$. Robots are points that move synchronously at time steps $t = 0, 1, \dots, K$ (planning horizon K). Motion is 4-connected (up, down, left, right) or stay. Robots share positions each step; communication desirability between robots is modeled by a weight that decreases with their Euclidean distance.

Map, Robots, and Positions

There are N robots indexed by $i = 1, \dots, N$. The position of robot i at time t is

$$p_{i,t} = (x_{i,t}, y_{i,t}), \quad x_{i,t} \in \{1, \dots, n\}, \quad y_{i,t} \in \{1, \dots, m\}.$$

Initial positions $p_{i,0}$ are given. The 4-neighborhood motion rule is

$$|x_{i,t} - x_{i,t-1}| + |y_{i,t} - y_{i,t-1}| \leq 1, \quad \forall i, \quad t = 1, \dots, K.$$

Coverage and Distance-Weighted Connectivity

Coverage variable for each cell:

$$C_{x,y} = \begin{cases} 1, & \text{if some robot stands on } (x, y) \text{ at any } t \in \{0:K\}, \\ 0, & \text{otherwise.} \end{cases}$$

Coverage updates via the visit indicator:

$$C_{x,y} \geq \mathbf{1}[(x_{i,t}, y_{i,t}) = (x, y)], \quad \forall (x, y), i, t.$$

Let the Euclidean distance between robots i and j at time t be

$$d_{ij,t} = \sqrt{(x_{i,t} - x_{j,t})^2 + (y_{i,t} - y_{j,t})^2}.$$

We define a *distance-weighted link quality* $W_{ij,t} \in [0, 1]$ that is inversely proportional to distance.

$$W_{ij,t} = \begin{cases} \frac{1}{1 + d_{ij,t}}, & d_{ij,t} \leq R_c \\ 0, & d_{ij,t} > R_c \end{cases}$$

(which preserves “inverse with distance” while ignoring very distant pairs.)

Objective Function

Maximize coverage and total distance-weighted connectivity across time:

$$\max f = \underbrace{\alpha \sum_{x=1}^n \sum_{y=1}^m C_{x,y}}_{\text{coverage}} + \underbrace{\beta \sum_{t=0}^K \sum_{1 \leq i < j \leq N} W_{ij,t}}_{\text{distance-weighted connectivity}},$$

with weights $\alpha, \beta > 0$.

Constraints

1. Map bounds:

$$1 \leq x_{i,t} \leq n, \quad 1 \leq y_{i,t} \leq m, \quad \forall i, t.$$

2. Motion (4-neighborhood + stay):

$$|x_{i,t} - x_{i,t-1}| + |y_{i,t} - y_{i,t-1}| \leq 1, \quad \forall i, t = 1, \dots, K.$$

3. Obstacle avoidance (online knowledge):

$$p_{i,t} \notin O_t, \quad \forall i, t, \quad \text{with } O_t \subseteq O_{t+1}.$$

4. Collision avoidance:

$$(x_{i,t}, y_{i,t}) \neq (x_{j,t}, y_{j,t}), \quad \forall i \neq j, t.$$

5. Coverage update:

$$C_{x,y} \geq \mathbf{1}[(x_{i,t}, y_{i,t}) = (x, y)], \quad \forall (x, y), i, t.$$

Decision Variables

The optimization chooses the next K positions for each robot:

$$\{ p_{i,t} = (x_{i,t}, y_{i,t}) \mid i = 1, \dots, N, t = 1, \dots, K \}.$$

Chapter 4

Results

Chapter 5

Conclusion

Chapter 6

Future Work

Appendix